refinement and evolution of the model in capturing the nuances of real-world financial markets.

Check Your Progress 3

1) What are some challenges and criticisms identified in empirical studies of the Arbitrage Pricing Theory (APT)?

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2) In the context of APT's comparison with other asset pricing models, what are the key considerations and criticisms?

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20.6 LET US SUM UP

The Arbitrage Pricing Theory (APT), introduced by Stephen Ross, serves as an alternative to the Capital Asset Pricing Model (CAPM) by incorporating multiple systematic risk factors influencing asset pricing. APT posits that the expected return on an asset is a linear function of various macroeconomic factors, with factor sensitivities (betas) determining the asset's response to each factor. Its applications extend to various financial markets, including equities, commodities, real assets, and fixed-income securities, with international considerations for cross-border asset pricing. Empirical studies have tested APT, identifying factors, assessing explanatory power, and addressing challenges like data limitations and model misspecification. Despite criticisms regarding assumptions and comparisons with other models, APT remains crucial for understanding asset pricing dynamics due to its flexibility and empirical testability. Ongoing research focuses on enhancing factor identification, incorporating dynamic models, considering behavioural aspects, and exploring non-linear relationships, highlighting APT's adaptability and potential for future advancements in the ever-evolving landscape of finance.

20.7 KEY WORDS

Arbitrage Pricing Theory (APT) : APT is an asset pricing model that offers an alternative to CAPM. It suggests that the expected return on an asset depends on multiple systematic risk factors, allowing for a more flexible framework compared to the single-factor CAPM.

Capital Asset Pricing Model (CAPM): CAPM is a financial model that establishes a relationship between the expected return on an investment and its systematic risk. It quantifies the expected return as a function of the risk-free rate, beta (systematic risk measure), and the market risk premium.

Efficient Market Hypothesis (EMH)	:	EMH asserts that financial markets efficiently incorporate all available information, making it impossible to consistently achieve higher-than-average returns through informed trading. It influences the assumptions of both CAPM and APT.
Factor Sensitivities ($Beta_{ij}$)	:	In APT, factor sensitivities represent how an asset responds to each specific systematic risk factor. Each beta ($beta_{ij}$) measures the asset's sensitivity to a particular macroeconomic factor.
Market Portfolio	:	The market portfolio in CAPM is a theoretical portfolio that includes all assets in the market, weighted by their market values. CAPM assumes that investors hold this portfolio along with the risk-free asset.
Market Risk Premium	:	The market risk premium is the excess return expected from the overall market compared to the risk-free rate. It is a key input in both CAPM and APT formulas, representing compensation for taking on systematic risk.
Rational Expectations	:	Rational expectations assume that individuals form their expectations about future asset prices based on all available information. Both CAPM and APT rely on the assumption of rational expectations for the proper functioning of markets and pricing models.
Risk-Free Rate	:	The risk-free rate represents the theoretical return on an investment with zero risk. It is typically approximated using the yield on government bonds, such as Treasury bills.
Systematic Risk (Beta, β)	:	Beta is a measure of an asset's sensitivity to changes in the market portfolio. A beta of 1 indicates the asset moves in line with the market, while a beta less than 1 implies lower volatility, and a beta greater than 1 indicates higher volatility.
Unsystematic Risk	:	Also known as specific or idiosyncratic risk, unsystematic risk is the risk that can be diversified away in a well-constructed portfolio. APT assumes that investors can eliminate unsystematic risk through proper portfolio diversification.

20.8 SOME USEFUL BOOKS/ARTICLES

Ross, Stephen A. Randolph W. Westerfield, and Jeffrey Jaffe (2019). *Corporate Finance: Core Principles and Applications*" McGraw Hill

Ross A.S, The Arbitrage Theory of Capital Asset Pricing, Journal of Economic Theory, Volume 13, Issue 3,1976, Pages 341-360, ISSN 0022-0531.

Jensen, Michael C. and Black, Fischer and Scholes, Myron S. and Scholes, Myron S., The Capital Asset Pricing Model: Some Empirical Tests, in Michael C. Jensen (ed), *Studies In The Theory Of Capital Markets*, Praeger Publishers Inc., 1972.

20.9 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) Single-Factor Model, homogeneous Expectations, Static Risk, Risk-Free Rate Assumption.

2) Perfect Markets, homogeneous Assets, normal Distribution of Returns, constant correlations, investor rationality.

Check Your Progress 2

1) Refer to Sub-section 20.3.1

2) $E(R_i)=R_f+\beta_{i1}\times\lambda_1+\beta_{i2}\times\lambda_2+\ldots+\beta_{ik}\times\lambda_k+\epsilon_i$

   Investors cannot consistently earn risk-free profits by exploiting price discrepancies between assets. Rationality is crucial for the proper functioning of arbitrage and for ensuring that asset prices align with the expectations implied by the model.

3) (CAPM) relies on a single systematic risk factor, which is market risk. (APT) allows for the consideration of multiple systematic risk factors.

Check Your Progress 3

1) Data Limitations, model misspecification, time-varying betas, factor redundancy, market efficiency concerns.

2) Refer to Sub-section 20.5.2

UNIT 21 PRICING OF DERIVATIVES

Structure

21.0 OBJECTIVES

After going through this unit, you will be able to:

- explain the concept of a derivative;
- describe the various types of derivatives;
- discuss various models of derivative pricing; and
- explain different models of derivative pricing.

21.1 INTRODUCTION

In recent years, derivative products such as futures and options have gained significant prominence in the Indian stock markets, playing crucial roles in price discovery, portfolio diversification, and risk hedging. Notably, the National Stock Exchange (NSE) has witnessed a remarkable surge in trading volumes within its derivative markets, surpassing the turnover observed in the spot markets.

The notional turnover (i.e. Sum of (the strike price of each contract traded today multiplied by the number of the underlying asset in each contract) of futures and options on the National Stock Exchange of India Ltd., recognized as the world's largest derivatives bourse in terms of the number of contracts traded, has experienced a notable increase over the past year. As of 2nd January, 2024, based on the 10-day moving average, the turnover reached \$5.04 trillion, marking a significant rise from \$2.5 trillion recorded at the close of 2022, according to information gathered by Bloomberg.

Given this substantial growth, understanding the intricacies of derivative instruments becomes increasingly vital for market participants. These financial tools continue to play a pivotal role not only in enhancing price discovery mechanisms but also in providing avenues for portfolio diversification and effective risk management. The persistently higher turnover in the derivative markets, as compared to the cash segment, underscores the significance of these instruments in the contemporary Indian financial landscape. You have already studied about basic concepts of derivatives in Unit 13 where we discussed about characteristics of derivatives and types of derivatives. In this unit, we will extend our discussion and examine various models of derivative pricing.

21.2 DERIVATIVES: BASIC CONCEPTS

We have discussed about derivatives and related concepts in Unit 13. We already know that "Derivative" implies that its value is not independent but rather "derived" from an underlying asset, which can range from financial instruments like securities and commodities to various other entities. Derivatives encompass contracts such as forwards, futures, options, or hybrids, with predetermined durations, linked to the value of specified real or financial assets or an index of securities. Referred to as "deferred delivery of deferred payment instruments," derivatives resemble securitized assets but lack the backing obligations from the original issuer of the underlying assets.

Derivative contracts have fixed expiry periods, typically ranging from 3 to 12 months, starting from the contract commencement date. The contract value depends on both the expiry period and the underlying asset's price. For instance, a farmer concerned about potential declines in wheat prices may enter into a derivative contract with a merchant, securing a set price (exercise price) for the crop at a future date. The derivative's value fluctuates based on the underlying asset's market value.

Commodity derivatives involve underlying assets like coffee, wheat, or precious stones, while financial derivatives involve assets such as debt instruments, currency, or equity shares. Derivatives can be exchange-traded or customized over-the-counter (OTC) based on standardization or negotiation with the involved parties.

To illustrate, a farmer anticipating a soybean production of 150 quintals in three months may either engage in an OTC derivatives contract with a food merchant or purchase a standardized wheat contract on a commodities exchange, leaving some quantity uncovered for price fluctuations. Exchange-traded derivatives offer advantages like lower transaction costs and reduced default risks.

The economic functions of derivatives markets include risk transfer from risk-averse to risk-oriented individuals, price discovery for future and current prices, stimulation of entrepreneurial activity, increased market trading volumes due to the participation of risk-averse individuals, and promotion of savings and long-term investments.

Check Your Progress 1

1) Explain the concept of a derivative.

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2) List the various functions of derivative markets.

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21.3 TYPES OF DERIVATIVES

In the realm of financial markets, a diverse array of derivatives is actively traded. We have discussed about types of derivatives in Unit 13. In this unit, we will revisit them briefly. Since these financial instruments play a significant role in financial market, we will focus on following derivatives in this unit:

1) Forward Contracts

2) Futures Contracts

3) Options

4) Swaps

21.3.1 Forward Contract

In financial markets, the ancient practice of Forward Contracts (forwards) has endured across various nations, including India, standing out as the most straightforward derivative. You must have already understood from Unit 13 that a forward contract is a bespoke **agreement between two entities, culminating in settlement on a predetermined future date at an agreed-upon current price.** This bilateral contract involves one party assuming a long position, committing to buy the underlying asset on a specified future date for a predetermined period. Simultaneously, the other party assumes a short position, agreeing to sell the asset on the same date at the same price.

For example: Let us consider a scenario where two parties enter into a contract to buy and sell 100 shares of Reliance at Rs 850 per share, two months from the contract date. The terms, including the product (Reliance shares), quantity (100 shares), price (Rs. 850 per share), and delivery time (2 months from the contract date), are predetermined.

However, forward contracts carry default risks. For instance, if the market price of Reliance shares significantly rises, the seller may prefer selling in the open market rather than honouring the contract, leading to a potential default. Similarly, if prices drop, the buyer may default, opting to purchase at the lower market rate. This mutual exposure to default risk is a drawback of customized forward contracts.

In India, forward markets, like those in the foreign exchange market, offer flexibility in negotiating terms such as price, quantity, quality, and delivery time and place. However, due to customization, they suffer from poor liquidity and credit risks.

To calculate payoffs from forward contracts, consider an example where two parties agree to exchange 500,000 barrels of crude oil at US $ 42.08 per barrel in three months. A forward contract involves variables like the underlier, notional amount (n), delivery price (k), and settlement date. The payoffs are linear, determined by the market value formula: $n(s - k)$. Here s stands for the price prevailing in the spot market. In a forward contract, the "underlier" refers to the asset or security that the contract is based on. It is the specific item or financial instrument that the buyer and seller agree to exchange at a future date. The underlier could be a commodity, currency, stock, bond, or any other tradable asset.

For example, if two parties enter into a forward contract to buy and sell 100 shares of Company X's stock at a predetermined price three months from now, the stock of Company X is the underlier in that forward contract. In essence, the underlier is the subject of the forward contract, and its value or performance will determine the financial outcomes and obligations of the contracting parties at the time of settlement.

Forward contracts may be cash-settled, meaning the underlier and payment don't physically exchange hands. Instead, a single payment is made based on the market value at settlement. The bid and offer forward prices, influenced by factors like the time value of money and market expectations, are crucial. The graph of forward prices for various maturities is known as a forward curve.

21.3.2 Futures

In financial markets, Futures Contracts represent legally binding agreements to buy or sell an underlying security on a predetermined future date. These contracts are organized and standardized concerning quantity, quality (for commodities), delivery time, and place, allowing for settlement on any future date. As you already know from Unit 13, the contract has a specified expiry date and settlement can occur either through the delivery of the underlying asset or in cash.

In contrast to Forward Contracts, Futures Contracts are typically traded on exchanges. The organized nature of these markets ensures high liquidity, mitigating the liquidity issues often present in the forward market. A clearing corporation or house in futures markets acts as the counterparty for all trades, providing an unconditional guarantee for settlement and maintaining financial integrity.

For instance, consider the scenario where parties A and B enter into a contract to buy and sell Reliance shares. If this contract is facilitated through an exchange, with a clearing corporation providing settlement guarantees, it is termed a futures contract.

The transition from a standardized and exchange-traded forward contract to a futures contract is evident, with futures offering advantages such as transparency and robust risk management. Settlement of futures contracts can occur through physical delivery of assets or in cash, with clear specifications outlined by the exchanges.

In a futures transaction, two parties negotiate through brokerage firms that hold a “seat” on the exchange for that particular contract. Legally structured as two contracts, each with the clearinghouse, this arrangement eliminates direct legal obligations between the parties. The margining process, involving daily profit or loss calculations, ensures ongoing settlement, effectively eliminating credit risk for futures.

Futures contracts can take various forms, including Index Futures and Stock Futures. Index Futures are based on an underlying index, cash-settled and deriving value from the index itself. On the other hand, Stock Futures involve specific stocks as underliers, with single stock futures being cash-settled.

21.3.3 Options

As we have discussed in Unit 13, Options is a contract, a subset of derivatives contracts that allows the buyer/holder the right (without the obligation) to buy/sell the underlying asset at a predetermined price within or at the end of a specified period. The buyer acquires this right from the seller, known as the writer, in exchange for a consideration called the premium. The seller is obligated to fulfil the terms of the contract when the buyer decides to exercise their right. Underlying assets for options can include securities, indices, or other financial instruments.

As per the Securities Contracts (Regulations) Act, 1956, options on securities are defined as contracts for the purchase or sale of the right to buy or sell securities in the future, encompassing various types like teji, mandi, galli, put, call, or put and call in securities. We have already discussed about Call and Put options in Unit 13 in details.

A Call option provides the right to buy, while a Put option provides the right to sell. American options can be exercised anytime on or before the expiry date, while European options are exercisable only on the expiry date. The agreed-upon price at which the option can be exercised is called the Strike price or Exercise price. Unlike forward or futures contracts, acquiring an option involves a cost known as the premium.

For instance, consider a European exercise, a three-month put option on 100,000 barrels of Brent oil with a strike price of US $45. This option gives the holder the right to sell 100,000 barrels of Brent oil at US $45 per barrel three months from today.

An analogy to buying a television illustrates the concept further. If Mr. A pays Rs. 200-300 to reserve a television for two days with the option to buy later, he is akin to an option buyer. If he finds a better deal elsewhere, he may choose to let the option expire, akin to not exercising the right. The Rs. 200-300 paid serves as the price of the option.

Options contracts can be settled through the delivery of the underlying asset or in cash. Cash settlement in options involves paying/receiving the difference between the strike price and the price of the underlying asset at the time of expiry or exercise/assignment.

Index options contracts, based on an index, provide the right but not the obligation to buy/sell the underlying index on expiry. These are generally European-style options and are cash-settled on expiry.

In options markets, the exercise price determines whether a call option is in the money (current price exceeds exercise price) or out of the money (current price is less than exercise price). The premium is the price paid by the buyer to the seller, and a covered option is written against an owned asset, while a naked option is written without owning the asset.

21.3.4 Swaps

In Unit 13, we discussed that a **swap** is a contractual agreement between two parties to exchange assets, financial obligations, or cash flows over a specified period at predetermined intervals. Various forms of swaps have emerged in the market, including interest rate swaps, coupon swaps, basis rate swaps, bond swaps, substitution swaps, Intermarket spread swaps, swaps with timing mismatches, swaps with options like payoffs, and currency swaps. Among these, interest rate swaps and currency swaps are the most commonly employed. We have discussed different forms of Swaps in Unit 13. The most prevalent type of swap is the “plain vanilla” interest rate swap. In this arrangement, a company commits to paying cash flows equivalent to interest at a predetermined fixed rate on a notional principal for a specified period. In return, it receives interest at a floating rate on the same notional principal for the same duration, often linked to the London Interbank Offer Rate (LIBOR).

An interest rate swap involves the exchange of one stream of interest obligations between two parties, with a specific maturity on a notional principal amount. The notional principal serves as a reference amount for interest calculations, with no actual exchange of principal occurring. Maturities for these transactions can range from less than a year to over 15 years, with the majority falling within the two to 10-year period.

On the other hand, a currency swap is a contractual agreement where foreign currencies are exchanged in the spot market, with a simultaneous commitment to reserve the transaction in the forward market. The exchange rate and timing of the forward market transaction are predetermined at the time of the swap. In essence, a currency swap involves swapping both principal and interest in one currency for equivalent amounts in another currency. Upon maturity, the principal amounts are swapped back.

For instance, a company facing exchange rate risk due to borrowing rupees at a fixed interest rate can mitigate this risk by entering a currency swap. The company sets up a contract to receive rupees at a fixed rate in exchange for dollars at either a fixed or floating interest rate. This strategy allows the company to manage both interest rate and exchange rate risks effectively.

The derivative market has achieved considerable success, attracting diverse traders due to its liquidity. **Traders in this market fall into three broad categories: hedgers, speculators, and arbitrageurs.**

- **Hedgers**: This group employs forwards, futures, and options to mitigate the risk associated with potential future movements in market variables. By utilizing derivative instruments, hedgers seek to safeguard themselves against adverse market fluctuations.

- **Speculators**: Traders in this category use derivatives to speculate on the future direction of market variables. Unlike hedgers who aim to reduce risk, speculators actively bet on market movements, seeking profit from price fluctuations.

- **Arbitrageurs**: Arbitrageurs engage in taking offsetting positions in two or more instruments simultaneously to lock in a profit. They capitalize on price differentials between related assets in different markets, exploiting temporary discrepancies for financial gain.

21.3.5 Put – Call Parity

In financial mathematics, put–call parity establishes a relationship between the prices of European call and put options with identical strike prices and expiry dates. This relationship relies on the absence of arbitrage in the market.

For example, considering stock options, let's denote the underlier value at expiration as S and the strike price as K. If S is less than or equal to K, the put option has a value of (K–S), and the share has a value of S. If S is greater than or equal to K, the put option has a value of 0, and the share has a value of S.

Considering two portfolios — one with a put option and a share, and the other with a call option and K bonds — the values at expiration are compared. The portfolios have equal values at any time t before expiration. This equality prevents arbitrage opportunities, establishing the following relationship between the instruments' values at time t:

$C(t)+K\times B(t,T)=P(t)+S(t)$

Where:

- $C(t)$ is the time-t value of the call
- $P(t)$ is the time-t value of the put
- $S(t)$ is the time-t value of the share
- K is the strike price
- $B(t, T)$ is the time-t value of a bond that pays at T.

Put–call parity implies:

1) **Equivalence of calls and puts:** Calls and puts can be used interchangeably in any delta-neutral portfolio. If d is the call's delta, then buying a call and selling d shares of stock is equivalent to buying a put and buying 1–d shares of stock. This equivalence is crucial in options trading.

   Parity of implied volatility: In the absence of dividends or other costs of carry, the implied volatility of calls and puts must be identical, ensuring consistency in market expectations.

In summary, understanding the diverse roles of hedgers, speculators, and arbitrageurs, along with the principles of put-call parity, contributes to a comprehensive grasp of derivative markets and their underlying mathematical relationships.

Check Your Progress 2

1) Explain what is meant by a forward contract.

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2) Explain the workings of a futures contract.